The Zero-Sum Ramsey Number of Two Disjoint Triangles Modulo 3 is 8

Research Note

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Abstract

For a graph G with $k \mid e(G)$, the zero-sum Ramsey number $R(G, \mathbb{Z}_k)$ is the smallest integer N such that every edge-weighting $c : E(K_N) \to \mathbb{Z}_k$ contains a copy of G whose edge sum is divisible by k. Addressing Problem 4 posed by Caro and Mifsud (*arXiv:2502.03864*), which asks whether $R(G, \mathbb{Z}_3) \leq |V(G)| + 8$, with equality if and only if $G = K_3$, we establish the exact value for the disjoint union of two triangles:

$$R(2K_3, \mathbb{Z}_3) = 8.$$

We prove analytically that $R(2K_3, \mathbb{Z}_3) \geq 8$ via an explicit vertex-potential construction on K_7, and show that the vertex-potential method cannot witness a value greater than 8. The matching upper bound $R(2K_3, \mathbb{Z}_3) \leq 8$ is established computationally by encoding the existence of a zero-sum-free weighting of K_8 as an exact CNF formula with 252 variables and 2,689 clauses, which is found to be unsatisfiable by a CDCL SAT solver. This demonstrates that the additive surplus parameter $\Delta(G) \coloneqq R(G, \mathbb{Z}_3) - |V(G)|$ collapses from $\Delta(K_3) = +8$ to $\Delta(2K_3) = +2$, matching the known upper bound for acyclic forests.

1 Introduction

Let $G = (V, E)$ be a finite simple graph, and let $\mathbb{Z}_k$ denote the cyclic group of order k. An edge-weighting of the complete graph K_N is a mapping $c : E(K_N) \to \mathbb{Z}_k$. A subgraph $H \subseteq K_N$ is *zero-sum* (modulo k) if

$$w(H) \coloneqq \sum_{e \in E(H)} c(e) \equiv 0 \pmod{k}.$$

The *zero-sum Ramsey number* $R(G, \mathbb{Z}_k)$ is the smallest integer N such that every edge-weighting of K_N over $\mathbb{Z}_k$ forces a zero-sum copy of G. A necessary condition for the existence of $R(G, \mathbb{Z}_k)$ is that $k \mid e(G)$; otherwise, the constant coloring $c(e) = 1$ yields no zero-sum subgraph.

The study of zero-sum graph problems was initiated by Bialostocki and Dierker [1] as an extension of the classical Erdős-Ginzburg-Ziv theorem [4]. In 1996, Caro [2] completely characterized $R(G, \mathbb{Z}_2)$ for all graphs with even edge counts.

Over $\mathbb{Z}_3$, the behavior of cyclic graphs contrasts sharply with that of forests. For any forest F on n vertices with $3 \mid e(F)$, Caro and Mifsud [3] established that $R(F, \mathbb{Z}_3) \leq n + 2$. However, for the single triangle K_3, classical evaluations show that $R(K_3, \mathbb{Z}_3) = 11 = |V(K_3)| + 8$ (see [1, 3]). This motivated the following conjecture:

Problem 1.1 (Caro & Mifsud [3], Problem 4)**.** Let G be a graph on n vertices such that $3 \mid e(G)$. Is it true that

$$R(G, \mathbb{Z}_3) \leq n + 8,$$

with equality if and only if $G = K_3$?

In this note, we settle the value for the disjoint union of two triangles $G = 2K_3$ $(n = 6, e = 6)$.

Theorem 1.2. *The zero-sum Ramsey number of two disjoint triangles modulo* 3 *is*

$$R(2K_3, \mathbb{Z}_3) = 8.$$

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2 The Lower Bound: $R(2K_3, \mathbb{Z}_3) \geq 8$

2.1 Analytical Construction on K_7

Proposition 2.1. $R(2K_3, \mathbb{Z}_3) \geq 8$.

Proof. We construct an edge-weighting of K_7 with no zero-sum copy of $2K_3$ using the vertex-potential method

$$c(uv) = a_u + a_v \pmod 3.$$

Assign potentials $a : V(K_7) \to \mathbb{Z}_3$ such that two vertices have potential 1 and five vertices have potential 2. Thus

$$(n_0, n_1, n_2) = (0, 2, 5).$$

The total potential sum is

$$S = \sum_{v \in V(K_7)} a_v = 2(1) + 5(2) = 12 \equiv 0 \pmod 3.$$

For any triangle $T = \{u, v, w\}$, its edge sum is

$$w(T) = (a_u + a_v) + (a_u + a_w) + (a_v + a_w) = 2(a_u + a_v + a_w) \equiv -(a_u + a_v + a_w) \pmod 3.$$

Any copy of $2K_3$ in K_7 spans six vertices and therefore omits exactly one vertex, say w. Hence the total weight of the two vertex-disjoint triangles T_1, T_2 is

$$w(T_1 \cup T_2) \equiv -\sum_{v \in V(T_1 \cup T_2)} a_v = -(S - a_w) \equiv a_w - S \equiv a_w \pmod 3.$$

Since $n_0 = 0$, every vertex $w \in V(K_7)$ satisfies $a_w \in \{1, 2\}$. Therefore

$$w(T_1 \cup T_2) \not\equiv 0 \pmod 3$$

for every copy of $2K_3$. Thus K_7 admits a zero-sum-free weighting, establishing $R(2K_3, \mathbb{Z}_3) \geq 8$. $\square$

2.2 The Potential Method Barrier on K_8

We prove that this algebraic technique cannot witness a bound greater than 8.

Lemma 2.2 (Potential Method Barrier)**.** *Every vertex-potential weighting* $c(uv) = a_u + a_v \pmod 3$ *on* K_8 *contains a zero-sum copy of* $2K_3$.

Proof. Let $a : V(K_8) \to \mathbb{Z}_3$ be an arbitrary vertex potential, and let $S \coloneqq \sum_{v \in V(K_8)} a_v \pmod 3$. Any copy of $2K_3$ in K_8 omits exactly two vertices, say $\{x, y\}$. Its combined weight satisfies:

$$w(T_1 \cup T_2) \equiv -(S - a_x - a_y) \equiv a_x + a_y - S \pmod 3.$$

Because any 6 vertices in K_8 induce a complete subgraph and can be partitioned into two disjoint triangles, the weighting avoids a zero-sum copy if and only if no two distinct vertices x, y satisfy:

$$a_x + a_y \equiv S \pmod 3. \tag{1}$$

Let $P = \{a_x + a_y \pmod 3 : 1 \le x < y \le 8\}$ be the set of realizable pair sums, and let $n_r = |\{v \in V(K_8) : a_v = r\}|$ for $r \in \{0, 1, 2\}$, with $n_0 + n_1 + n_2 = 8$.

For any residue $r \in \mathbb{Z}_3$, the sum $u + v \equiv r \pmod 3$ is realized either by:

1. Monochromatic pairs ($u = v$): requires $2u \equiv r \implies u \equiv -r \pmod 3$, realizable if $n_{-r} \ge 2$.
2. Heterochromatic pairs ($u \ne v$): requires $\{u, v\} = \mathbb{Z}_3 \setminus \{-r\}$, realizable if $\min(n_u, n_v) \ge 1$.

Thus, a residue r fails to appear in P ($r \notin P$) if and only if:

$$n_{-r} \le 1 \quad \text{and} \quad \min(n_u, n_v) = 0 \quad \text{where } \{u, v\} = \mathbb{Z}_3 \setminus \{-r\}.$$

This requires that at most one copy of $-r$ is present and at least one other residue is completely absent. Because $n_0 + n_1 + n_2 = 8$, the remaining non-absent residue must have multiplicity at least $8 - 1 = 7$.

Thus $P \ne \mathbb{Z}_3$ can occur only when the frequency vector (n_0, n_1, n_2) is, up to permutation, either $(8, 0, 0)$ or $(7, 1, 0)$. We check these remaining configurations directly:

- $(8, 0, 0) \implies S \equiv 0$, and $P = \{0 + 0\} = \{0\}$, so $S \in P$.
- $(0, 8, 0) \implies S \equiv 8(1) \equiv 2$, and $P = \{1 + 1\} = \{2\}$, so $S \in P$.
- $(0, 0, 8) \implies S \equiv 8(2) \equiv 1$, and $P = \{2 + 2\} = \{1\}$, so $S \in P$.
- $(7, 1, 0) \implies S \equiv 1$, and $P = \{0 + 0, 0 + 1\} = \{0, 1\}$, so $S \in P$.
- $(7, 0, 1) \implies S \equiv 2$, and $P = \{0 + 0, 0 + 2\} = \{0, 2\}$, so $S \in P$.
- $(1, 7, 0) \implies S \equiv 7 \equiv 1$, and $P = \{1 + 1, 1 + 0\} = \{2, 1\}$, so $S \in P$.
- $(0, 7, 1) \implies S \equiv 7 + 2 \equiv 0$, and $P = \{1 + 1, 1 + 2\} = \{2, 0\}$, so $S \in P$.
- $(1, 0, 7) \implies S \equiv 14 \equiv 2$, and $P = \{2 + 2, 2 + 0\} = \{1, 2\}$, so $S \in P$.
- $(0, 1, 7) \implies S \equiv 1 + 14 \equiv 0$, and $P = \{2 + 2, 2 + 1\} = \{1, 0\}$, so $S \in P$.

In every configuration, $S \in P$. Therefore, every potential weighting on K_8 contains a zero-sum $2K_3$. $\square$

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3 The Upper Bound: $R(2K_3, \mathbb{Z}_3) \leq 8$

To determine whether non-potential weightings can avoid zero-sum copies on K_8, we encode the existence of a zero-sum-free edge-weighting as a Propositional Satisfiability (SAT) instance in Conjunctive Normal Form (CNF).

Proposition 3.1. *Every edge-weighting $c : E(K_8) \to \mathbb{Z}_3$ contains a zero-sum copy of $2K_3$. Consequently, $R(2K_3, \mathbb{Z}_3) \leq 8$.*

Proof. We encode the existence of a zero-sum-free edge-weighting of K_8 as a CNF formula with 252 variables and 2,689 clauses:

1. **Variables:**

   - For each edge $e \in E(K_8)$ ($\binom{8}{2} = 28$ edges), three boolean variables $x_{e,0}, x_{e,1}, x_{e,2}$ encode its assigned weight in $\mathbb{Z}_3$. Total: $28 \times 3 = 84$ variables.
   - For each triangle $T \in \binom{V(K_8)}{3}$ ($\binom{8}{3} = 56$ triangles), three boolean variables $r_{T,0}, r_{T,1}, r_{T,2}$ encode its residue modulo 3. Total: $56 \times 3 = 168$ variables.

   Total variables: $84 + 168 = 252$.

2. **Clauses:**

   - *Edge exact-one constraints:* For each edge, exactly one weight in $\mathbb{Z}_3$ is selected (1 at-least-one clause and 3 at-most-one clauses): $28 \times 4 = 112$ clauses.
   - *Triangle exact-one constraints:* For each triangle, exactly one residue in $\mathbb{Z}_3$ is selected: $56 \times 4 = 224$ clauses.
   - *Triangle definitions:* For each triangle T with edges e_1, e_2, e_3 and every triple $(c_1, c_2, c_3) \in \{0, 1, 2\}^3$, we include the implication clause:

     $$\neg x_{e_1,c_1} \vee \neg x_{e_2,c_2} \vee \neg x_{e_3,c_3} \vee r_{T,(c_1+c_2+c_3 \bmod 3)}.$$

     This contributes $56 \times 27 = 1{,}512$ clauses.
   - *Zero-sum avoidance:* There are

     $$\frac{1}{2}\binom{8}{3}\binom{5}{3} = 280$$

     unordered pairs of vertex-disjoint triangles $\{T_1, T_2\}$. For each such pair, the three zero-sum residue combinations $(0, 0), (1, 2), (2, 1)$ are forbidden via:

     $$(\neg r_{T_1,0} \vee \neg r_{T_2,0}), \quad (\neg r_{T_1,1} \vee \neg r_{T_2,2}), \quad (\neg r_{T_1,2} \vee \neg r_{T_2,1}).$$

     This contributes $280 \times 3 = 840$ clauses.
   - *Symmetry breaking:* Adding a constant $a \in \mathbb{Z}_3$ to every edge alters the total weight of any 6-edge copy of $2K_3$ by $6a \equiv 0 \pmod 3$. Thus, fixing $x_{(0,1),0} = \text{True}$ preserves satisfiability without loss of generality (1 clause).

   Total clause count:

   $$112 + 224 + 1{,}512 + 840 + 1 = 2{,}689.$$

This exact 2,689-clause instance was evaluated using standard CDCL SAT solvers (including Z3 and CaDiCaL). The solvers determine the instance to be **UNSATISFIABLE** in under 0.2 seconds, establishing that no zero-sum-free edge-weighting of K_8 exists. Hence

$$R(2K_3, \mathbb{Z}_3) \leq 8.$$

The CNF formula and verification code are made available in an open reproducibility repository.[1] □

Combining Proposition 2.1 and Proposition 3.1 completes the proof of Theorem 1.2.
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4 Discussion and Concluding Remarks

Theorem 1.2 provides a sharp resolution of the two-component triangle case of Problem 4 of Caro and Mifsud [3]. Defining

$$\Delta(G) \coloneqq R(G, \mathbb{Z}_3) - |V(G)|,$$

we have

$$\Delta(K_3) = 8, \qquad \Delta(2K_3) = 2.$$

Thus, although the single triangle attains the larger +8 offset, adding a second disjoint triangle reduces the surplus to +2, matching the known upper bound for forests ($R(F, \mathbb{Z}_3) \leq |V(F)| + 2$).

This suggests that large surpluses over $\mathbb{Z}_3$ may be associated primarily with isolated odd cycles. We therefore propose the following conjecture:

Conjecture 4.1. *For any integer $s \geq 2$, the zero-sum Ramsey number of s disjoint triangles satisfies:*

$$R(sK_3, \mathbb{Z}_3) = 3s + 2.$$

In particular, $\Delta(sK_3) = 2$ for all $s \geq 2$.

References

[1] A. Bialostocki and P. Dierker, *Zero-sum Ramsey theorems*, Congr. Numer. **89** (1992), 193–198.

[2] Y. Caro, *Zero-sum problems—a survey*, Discrete Math. **152** (1996), no. 1-3, 93–113.

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[4] P. Erdős, A. Ginzburg, and A. Ziv, *Theorem in the additive number theory*, Bull. Res. Council Israel Sect. F **10** (1961), 41–43.

[1] https://github.com/vegaorion/zero-sum-ramsey-2k3